

iPhone Software Engineering Assessment 1 Report

App name: Itineria
Category: Travel, Planning

Student names

Phan Duc Anh Nguyen (S3915181)
Daniel La (S3909873)

Tutorial Session

Wednesday 5:30PM – 7:30PM
Tutor: James Dale

Instructor

Shekhar Kalra

RMIT University
COSC2472





Welcome to **Itineria**, a travel app designed to enhance and simplify the travel planning experience. This app is being developed by group **iPSE_22**, consist of members Daniel La (**S3909873**) and Phan Duc Anh Nguyen (**S3915181**). This report describes the development process of our platform. You can explore our work and contribution through these links.

GitHub Repository

<https://github.com/rmit-iPSE-2024-s2/a1-S3909873-S3915181.git>

Figma (Wireframes Prototype):

<https://www.figma.com/design/N2yovaok9L6B7kQxfZ9L0g/iPhone-Engineering-Prototype-Design?node-id=0-1&t=rtlu6xyzjy146XP7-1>

Table of Contents

<i>The Purpose of Itineria</i>	3
<i>Detailed Review of 4 Related Travel Apps</i>	3
Stippl App.....	3
Wanderlog App	5
Sygic Travel App	6
Polarsteps App	8
<i>Human Interface Guideline</i>	9
App Icons	9
Colour	9
Maps	9
Settings	10
Tab View	10
<i>Wireframes</i>	10
Profile Screen	10
Create Itinerary Screen	10
Map Screen.....	11
Edit Profile Screen	11
Settings Screen.....	11
<i>Data Requirements and API</i>	11
Data requirement.....	11
User	11
Itinerary.....	12
Destination	12
Activities	12
Map.....	13



Sharing and Collaboration Data	13
Data model	13
User	13
Itinerary	13
ItineraryItem	14
Destination	14
Activity	14
Map	15
Shared Itinerary	15
API	15
Itinerary & Destination Data	15
Map & Navigation Data	16
Activities Data	16
References	17

The Purpose of Itineria

Our travel app is specifically designed for travellers, offering a comprehensive travel planner that simplifies the entire travel planning process. The app's main task is for users to create detailed itineraries with integrated map functionalities. Users can select their desired destinations, along with adding notes, descriptions and pictures, to create an informative travel plan or memory. The itineraries can also be shared with friends and family, enabling users to share their memorable experiences with others. By incorporating this core feature within our app, we aim to address common travel challenges, by helping users stay well on top, organised, informed and connected throughout their journeys, ultimately making their travel experience more enjoyable and stress-free.

Detailed Review of 4 Related Travel Apps

Stippl App

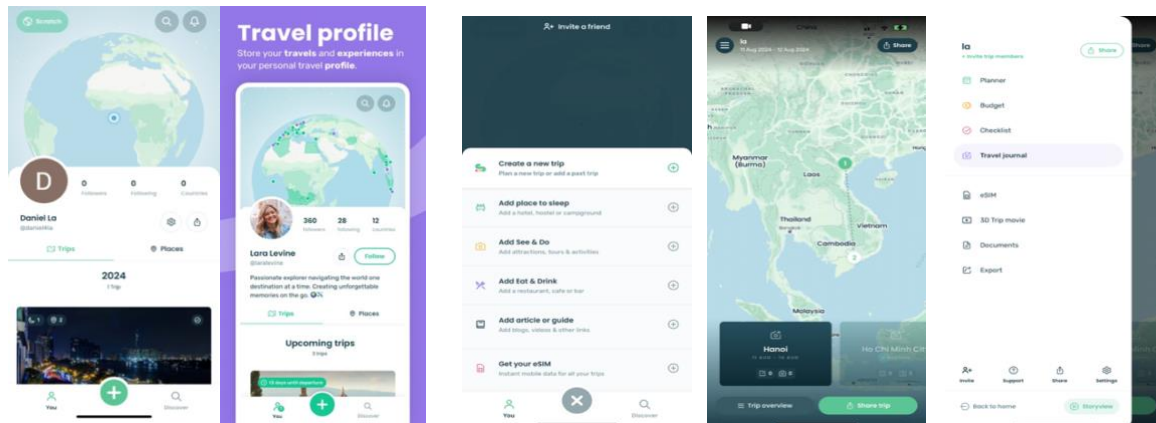
Stippl is a travel app designed to help users streamline their trip planning by offering comprehensive itinerary management and other features. After a detailed review of other travel-related apps, Stippl closely aligns with the travel app we aim to develop. The app facilitates effective creation and organisation of travel itineraries, allowing users to plan their activities, schedules, destinations, as well as sharing their experiences with others. Moreover, we were impressed by its aesthetic, utilising minimalistic design and light-coloured contrast throughout the app, which we plan to adopt a similar approach.

A downside of the Stippl app is that it can be overwhelming for users who may lack travel knowledge or experience with iPhone applications. The app contains vast range of travel features which were not well explained during and after the registration process, which can lead to users being too confused or overwhelmed. Even after reviewing the app, there are still many minor features that we did not know existed.

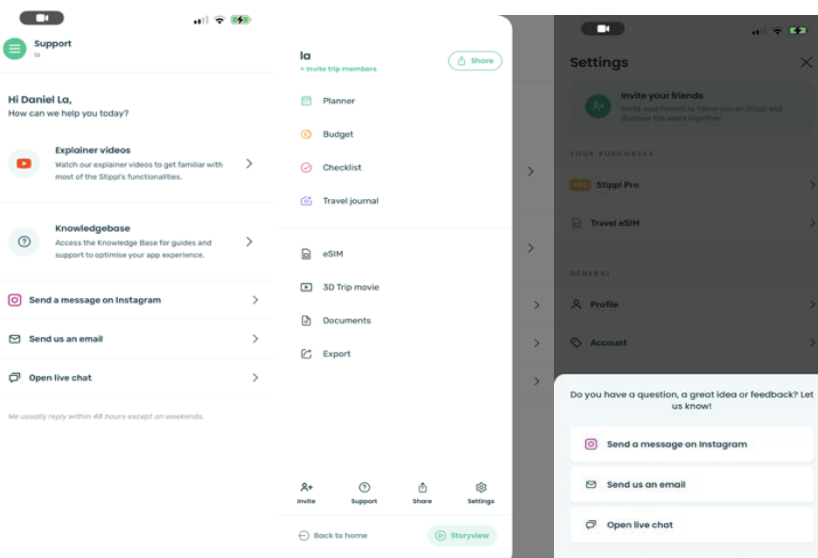


Design: Great contrast, simple and effective

Features: Many features

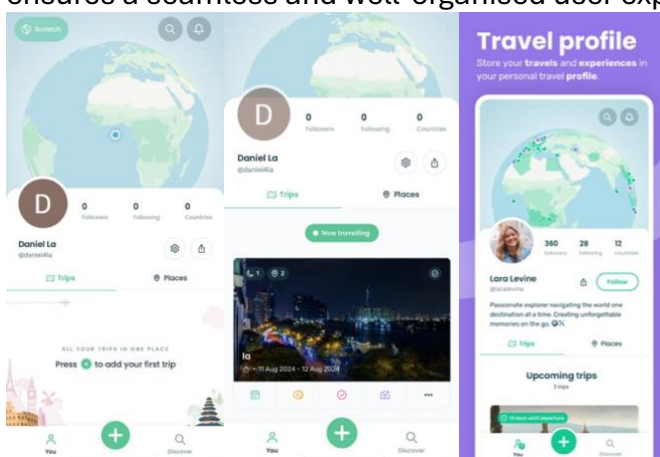


Stippl Issues: Multiple different settings and support pages. Can overwhelm users.



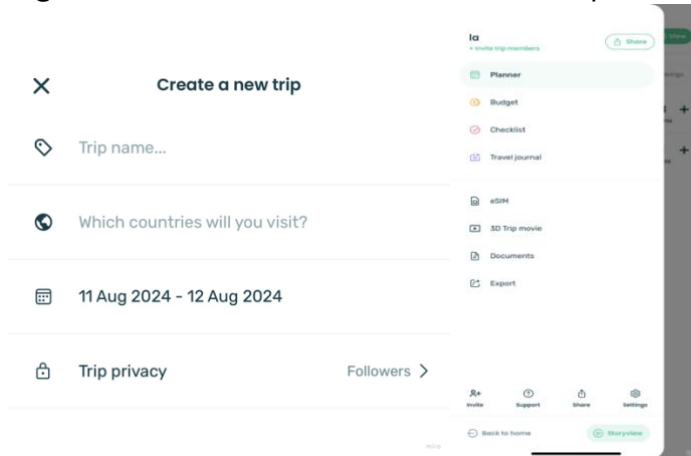
Here are a list of potential designs and features we plan to incorporate into our app:

- **Profile:** Stippl offers a clean and organised layout for users' profile. This design ensures a seamless and well-organised user experience.





- **Creating a trip/itinerary:** Stippl's approach in trip creation and itinerary management offers a straightforward process, allowing users to freely plan, organise and customise their own travel experience.

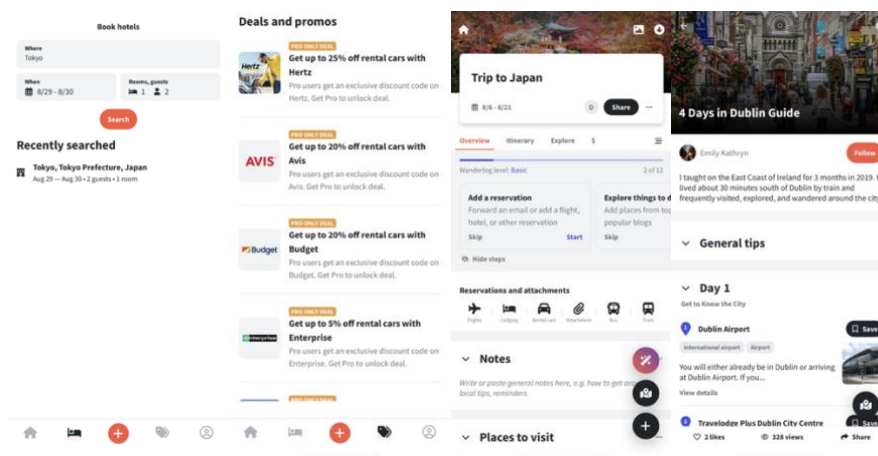


Wanderlog App

Wanderlog is another travel app with similar features to Stippl, which focuses on travel and itinerary planning. What separates Wanderlog compared to other travel apps is that the app focuses on sharing itineraries, experiences, and travel guide with a shared community. It provides public travel guides for users planning their destinations, supports itinerary management, offers promotions and redirects users to book hotels in their desired locations.

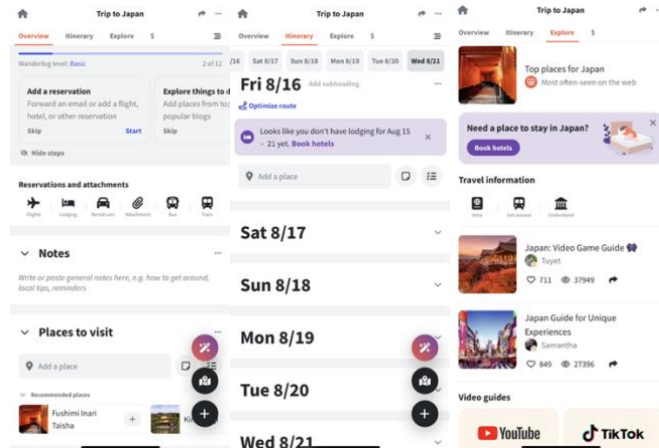
While Wanderlog may demonstrate that it excels in offering great features, app designs and vibrant coloured contrast, some areas of the app can feel overwhelming and clustered with information and text. This is prominently noticeable within the trip plan feature. The itinerary design lacks refinement, which can cause creating, organising and managing trips less intuitive, overall reducing user experience.

Wanderlog Features: Vast amount of features



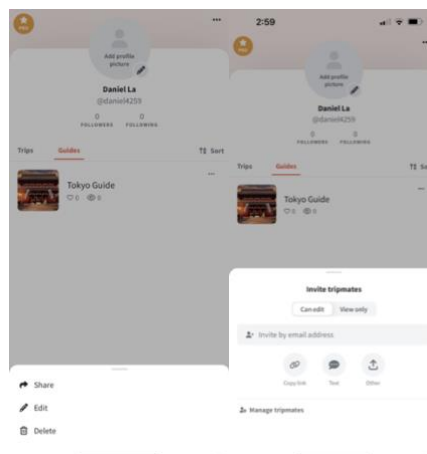


Wanderlog Issues: Planning is too overwhelming and clustered. Text fonts are not sized well.



Here is a potential feature we plan to incorporate into our app:

- **Sharing Trips:** Utilising ellipses on itineraries helps users recognise quick accessible functionalities such as share, edit and delete. This approach provides multiple functionalities quickly, as well as not overwhelming the user with its design.

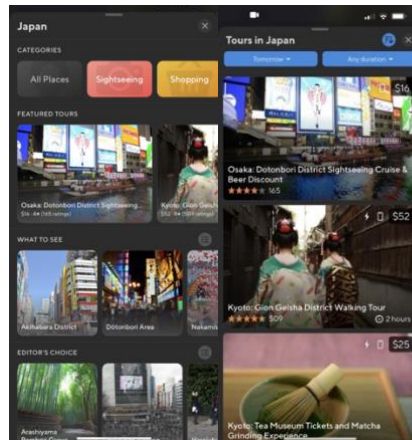


Sygic Travel App

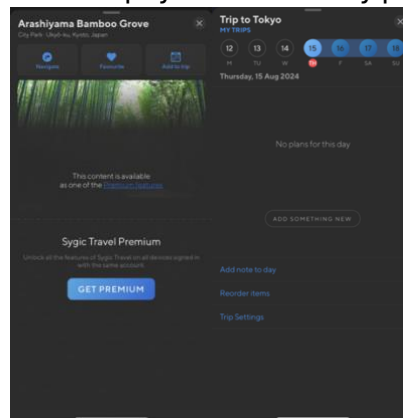
Sygic Travel App has offered valuable insights on developing the travel map feature for our app. Sygic was the most unique as its core features such as travel planning, itinerary management and many more, are centred around the map itself. Its distinct elements are its comprehensive guide, offering a detailed overview of hotspots, tourist attractions, and hotels. Additionally, the convenience of being able to book tours and hotels within the app, allows planning efficiency and organisation. While the Sygic Travel app offers many unique features, some of the content, including detailed guides on areas, is locked behind a payment option. This may limit the apps utility as many users prefer free access and would resort to other travel apps. Additionally, planning a trips itinerary is not as comprehensive as others.



Sygic Features: Unique features for a travel app

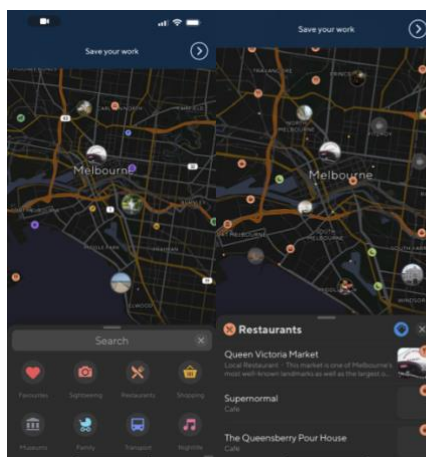


Sygic Issues: Content locked behind payment. Itinerary planning is not comprehensive.



Here is a potential design and feature we plan to incorporate into our app:

- **Map and Search Engine:** The map feature is Sygic's central component that provides a high level of interactive experience for users. It highlights key interests around the area, such as restaurant, tourist attractions, accommodation and many more.

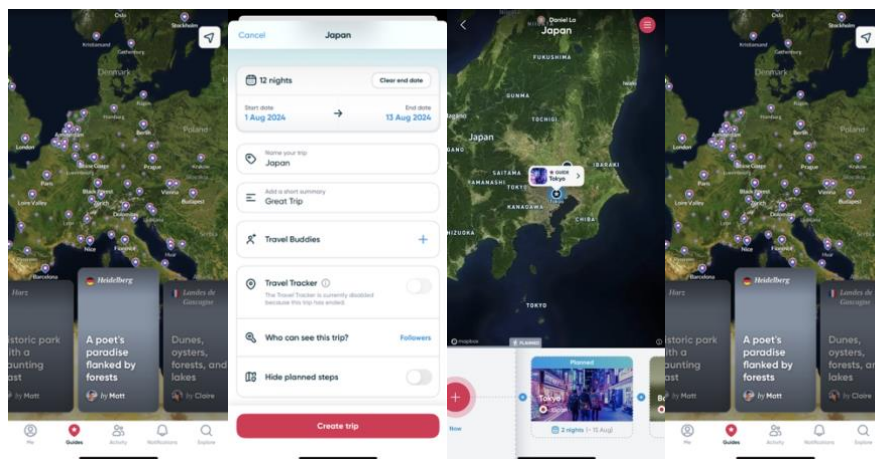




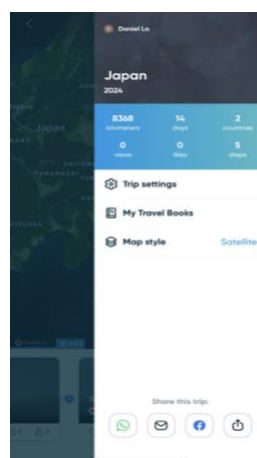
Polarsteps App

Polarsteps is a travel tracking app that has taken a different approach to other travel apps by allowing users to document their journey in real time, essentially creating a visual diary. The app tracks user location and records their trip, mapping out their travels to provide a comprehensive view of their journey. Users also have the option to share their adventures with others. Additionally, many other users share their own travel guides which offers detailed insights into tourist attractions, activities, favourites and many more, assisting others on their travel journey. While Polarsteps provides users with a great way to document and share their travel experiences, it falls short compared to other travel apps in terms of itinerary planning. The app's purpose is more to track and document users travel experiences rather than organising a detailed trip which many others would prefer. As a result, users may resort to other travel apps that include both itinerary documentations, as well as comprehensive planning.

Polarsteps Features: Amazing documentary features



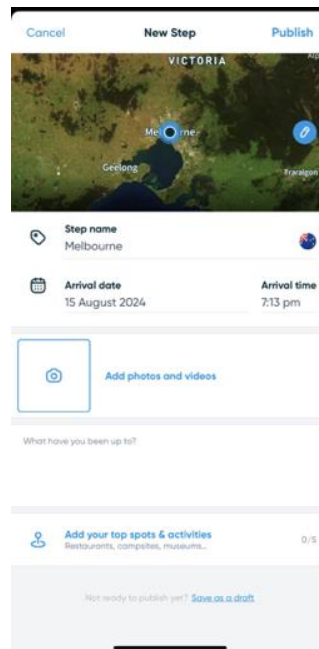
Polarsteps Issue: Not great for itinerary planning





Here is a potential design we plan to incorporate into our app:

Itinerary Layout: Polarsteps provides an organised itinerary layout, which is ideal for our app's design. This can be used as a reference when developing our app.



Human Interface Guideline

App Icons

By referring to the Human Interface Guidelines, we are going with a simple design with concept that can capture the essence of our app with no text included. Our app uses a simple vector image of a globe with three main colours (white, green, blue) which are also the theme colour of our application. We also designed the app in a rounded corner square shape as suggested by HIG. The icon has great contrast ratio, easy to see and high recognizability.



Colour

Our app would be using minimal colours (3 colours: white, blue and green) as in a non-game app, overuse of colour can make communication less clear and can distract people. Our app is going to have dark mode, and we are going to pick colours that can work well with both modes, so we choose white, blue and green which all work well with both dark and light mode.

Maps

Being a travel app, maps are a crucial feature. A standard style is chosen as it presents a fully saturated version. A search filter is implemented to help people find the app's content easily. Moreover, apple logo and legal link are highly visible to users as suggested by HIG.





Settings

For the settings tab, we minimize the number of setting options as more options call for confusion and un-friendly interface. The settings tab only has 3 setting items: change password, privacy & policy and notification.

Tab View

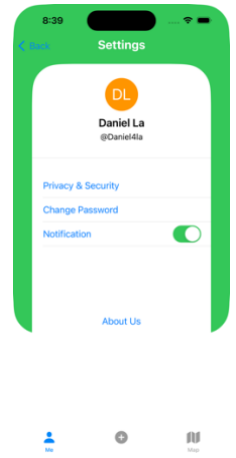
We designed a tab view below for easy navigation. There are three tabs and to avoid layout issues and being overwhelming. Each tab is also provided with labels as suggested by HIG

Wireframes

Figma Link: <https://www.figma.com/design/N2yovaok9L6B7kQxfZ9L0g/iPhone-Engineering-Prototype-Design?node-id=0-1&t=rtlu6xyzjy146XP7-1>

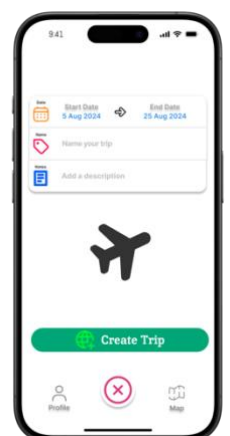
Profile Screen

The profile screen serves as the user's personal page, where they can view and update their profile details, including name, username, and profile picture. Users can also manage their settings from this screen. It provides access to itineraries they've created and their favorited trips. The workflow starts when users register, log in, or open the app, landing them on the Profile Screen by default. From here, they can effortlessly navigate to other areas of the app using the navigation bar.



Create Itinerary Screen

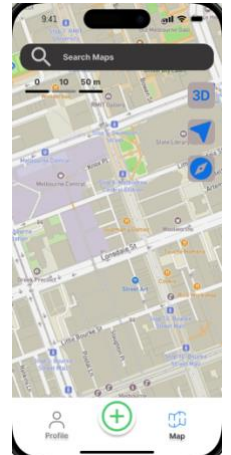
The Itinerary Creation Screen can be accessed using the navigation bar. This page enables users to enter trip details, including travel dates, a trip title, and a brief description. Once these details are provided, users can create the trip and start planning their itinerary. They can also use the navigation bar to access other areas of the app.





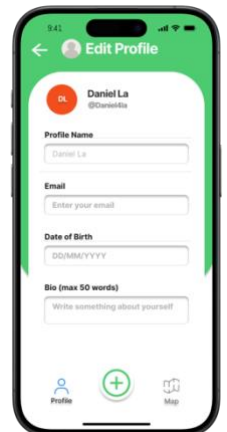
Map Screen

The map screen can be accessed using the navigation bar. This page offers an interactive map where users can search for their desired destinations. The workflow involves users interacting with the map to explore various locations as well as utilising inbuilt map features such as the search bar, 3D view, compass and user location. By selecting their intended destinations, users can add the location to their itinerary planner.



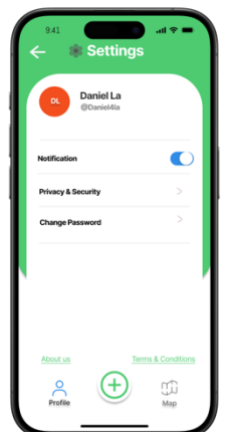
Edit Profile Screen

The edit profile screen enables users to update their personal information such as name, email, date of birth and bio. This page can be accessed from the profile screen. The workflow involves navigating from the profile page and updating information with users desired changes. This ensures users can personalise their profile, tailoring the app to their preference.



Settings Screen

The setting screen allows users to manage and customize app configurations to fit their preferences. This page can be accessed from the profile screen. The workflow involves navigating from the profile page, where users can adjust notification settings, manage privacy options, and update account password. This functionality enables users to tailor the app to their specific needs and preferences.



Data Requirements and API

Data requirement

User

User Profile Information

- UserID: Unique ID for each user



- Name: Full name
- Email: Email address for login and notification
- ProfilePicture: URL or binary data of the user's profile picture.
- Other essential personal information such as age, bio, introduction, etc.

Network

- FriendIDs: List of friends or connections.

Itinerary

Itinerary Information

- ItineraryID: Unique identifier for each itinerary.
- CreatorID: Reference to the user who created the itinerary.
- Title: Title of the itinerary
- StartDate: Start date of the trip.
- EndDate: End date of the trip.
- NumberOfActivities: number of activities included in the itinerary
- NumberOfDestinations: number of destinations included in the itinerary
- SharedWith: List of users with whom the itinerary is shared.
- CreatedDate: Date the itinerary was created.

Itinerary Items

- ItemID: Unique identifier for each item in the itinerary.
- ItineraryID: Reference to the itinerary it belongs to.
- DestinationID: Reference to the destination.
- ActivityID: Reference to the activity
- Title: Title of the item
- Description: Description about the item
- Notes: Additional note
- PlannedDate: Date of the planned activity.
- PlannedTime: Time of the planned activity.

Destination

- **DestinationID**: Unique identifier for each destination.
- **ActivityID**: Reference to the activity happening at this destination.
- **Country**: Name of the country.
- **City**: Name of the city or region.
- **Coordinates**: The coordinates of the destination.
- **GoogleMapURL**: Embedded Google Map URL for easy navigation.
- **Description**: Description or summary of the destination.

Activities

- **ActivityID**: Unique identifier for each destination.
- **DestinationID**: Reference to the destination where the activity happens.
- **Duration**: Estimate time to complete the activity.



- **Description:** Description or summary of the activity.

Map

- MapID: Unique identifier for map-related data.
- ItineraryID: Reference to the associated itinerary.
- RouteDetails: Details of the route (e.g., start and end coordinates).
- POIs: List of Point of Interests along the routes

Sharing and Collaboration Data

Shared Itineraries

- ShareID: Unique identifier for shared instances.
- ItineraryID: Reference to the shared itinerary.
- SharedBy: UserID of the person who shared the itinerary.
- SharedWith: UserID of the recipient.
- ShareDate: Date the itinerary was shared.

Data model

User

Attributes

- `UserID` (Primary Key)
- `Name`
- `Email`
- `ProfilePicture`
- `Age`
- `Bio`

Relationships

- A user can have multiple friends (Self-referencing Many-to-Many relationship).
- A user can create multiple itineraries (One-to-Many relationship with Itinerary).

Itinerary

Attributes

- `ItineraryID` (Primary Key)
- `CreatorID` (Foreign Key)
- `Title`
- `StartDate`
- `EndDate`
- `NumberOfActivities`
- `NumberOfDestinations`
- `SharedWith` (List of UserIDs)
- `CreatedDate`



Relationships

- An itinerary can contain multiple itinerary items (One-to-Many relationship with ItineraryItem).
- An itinerary can be shared with multiple users (Many-to-Many relationship with User).

ItineraryItem

Attributes

- `ItemID` (Primary Key)
- `ItineraryID` (Foreign Key)
- `DestinationID` (Foreign Key)
- `ActivityID` (Foreign Key)
- `Title`
- `Description`
- `Notes`
- `PlannedDate`
- `PlannedTime`

Relationships

- An itinerary item belongs to a single itinerary (Many-to-One relationship with Itinerary).
- An itinerary item is associated with a single destination (Many-to-One relationship with Destination).
- An itinerary item is associated with a single activity (Many-to-One relationship with Activity).

Destination

Attributes

- `DestinationID` (Primary Key)
- `Country`
- `City`
- `Coordinates`
- `GoogleMapURL`
- `Description`

Relationships

- A destination can have multiple activities (One-to-Many relationship with Activity).
- A destination can be linked to multiple itinerary items (One-to-Many relationship with ItineraryItem).

Activity

Attributes

- `ActivityID` (Primary Key)
- `DestinationID` (Foreign Key)
- `Duration`



- `Description`

Relationships

- An activity is linked to a single destination (Many-to-One relationship with Destination).
- An activity can be part of multiple itinerary items (One-to-Many relationship with ItineraryItem).

Map

Attributes

- `MapID` (Primary Key)
- `ItineraryID` (Foreign Key)
- `RouteDetails`
- `POIs`

Shared Itinerary

Attributes

- `ShareID` (Primary Key)
- `ItineraryID` (Foreign Key)
- `SharedBy` (Foreign Key, UserID)
- `SharedWith` (Foreign Key, UserID)
- `ShareDate`

Relationships

- A shared itinerary is linked to a single itinerary (Many-to-One relationship with Itinerary).
- A shared itinerary is associated with two users (SharedBy and SharedWith).

API

There's a few free API that we can implement to provide data for our app.

Itinerary & Destination Data

- **OpenWeatherMap API:**
 - **Purpose:** Provides current weather data for destinations.
 - **Usage:** To display weather information for destinations in itineraries.
- **Geonames API:**
 - **Purpose:** Provides detailed geographical data like cities, countries, coordinates, and more.
 - **Usage:** To retrieve destination information, including coordinates.
- **Google Places API:**
 - **Purpose:** Provides information on destinations and points of interest.
 - **Usage:** To get detailed data about destinations, such as addresses, coordinates, and photos.



Map & Navigation Data

- **Mapbox API:**
 - **Purpose:** Provides map and navigation data, including routes and POIs.
 - **Usage:** To integrate maps and provide route details within the app.
- **Here Maps API:**
 - **Purpose:** Provides detailed map, routing, and location data.
 - **Usage:** For map displays and route calculations in itineraries.

Activities Data

- **Eventbrite API:**
 - **Purpose:** Provides information about events happening in various locations.
 - **Usage:** To populate activities in itineraries with local events.



References

- AppStuff (27th June 2023) 'AMAZING NEW SwiftUI MapKit Features | iOS 17 | WWDC23' [video], AppStuff, YouTube Website, accessed 23rd August 2024. https://www.youtube.com/watch?v=gy6rp_pJmbo&t=247s
- Daniel Budd (31st December 2021) 'Setting an Image as a Button - 11 - Introduction to SwiftUI' [video], Daniel Budd, YouTube Website, accessed 29th August 2024. <https://www.youtube.com/watch?v=4MuzkObYmlo>
- RMIT Creds (2024) 'Example 9- Custom Layout' [code-archive], RMIT University, Melbourne.
- Core SwiftUI: Animation & Design (16th Decembre 2022) 'Using ChatGPT: How to Create a Spinning Rainbow SwiftUI Animation Effortlessly' [video], Core SwiftUI: Animation & Design, YouTube Website, accessed 29th August 2024. <https://www.youtube.com/watch?v=tlBzsDd3vc0>
- Wang H (2022) 20. SwiftUI: How to scale and rotate an image view, Fokswang Website, accessed 29th August 2024. <https://www.fokswang.com/post/20-swiftui-how-to-scale-and-rotate-an-image-view>
- Indently (22nd October 2021) 'How to create a Bottom Navigation Bar with TabView in Xcode (SwiftUI / iOS)' [video], Indently, Youtube Website, accessed 19th August 2024. <https://www.youtube.com/watch?v=TgvYFfCjDMo&t=213s>